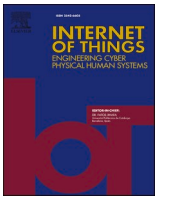
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Environmental noise monitoring using distributed hierarchical wireless acoustic sensor network

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ABSTRACT

Acoustic noise pollution is one of many problems people face as cities grow. Long-term noise exposure can result in a series of physical and mental health diseases that are highly harmful to foetuses and newborns. Hence, many IoT-based wireless sensor network systems have been proposed for automated monitoring for long-term operation. However, these systems suffer from weaknesses in functionality, power consumption, costs, and scalability, which hinder large-scale deployment. In this study, we propose a distributed hierarchical wireless acoustic sensor network for environmental noise monitoring to do sound classification and A-weighted sound-pressure-level measurement to address the shortcomings of existing systems. A series of tests and comparisons are performed in diagnosing the performance with respect to recording continuity, packet loss, recording quality, accuracy on A-weighted sound pressure level calculations, and costs. Results show that this proposed network structure is feasible as a part of hardware implementation in a large-scale, low-cost, and high-scalable environmental noise monitoring system to classify sound.

1. Introduction

Sound is closely related to people as a fundamental approach for humans to obtain information from the outside world. Although people have entered the information age and improved their quality of life in various ways, noise pollution is still one of the major issues people face [1,2]. Noise, which refers to unwanted sound with a high sound pressure level (SPL), should be handled seriously as it harms mental and physical health. Long-term noise exposure can result in hearing loss, mental health diseases, memory impairment, raised blood pressure, cardiovascular diseases, sleep disorders, decreased cognitive function, and more illnesses [2,3,4,5,6]. The incidence of diseases such as hypertension and myocardial infarction is significantly increased with a noise level in a range of 45–54 dBA during the daytime, and adverse effects on health can be caused by 40–42 dBA noise at night [7]. Foetuses and newborns are vulnerable to noise hazards. The Committee on Environmental Health indicates that exposure to high-level noise during pregnancy leads to high-frequency hearing loss in newborns and may be the reason for prematurity and intrauterine growth retardation. Therefore, effective noise-intervention strategies are necessary to minimise the health impact of noise.

During the development of noise-intervention strategies, it is critical to not only calculate noise SPLs but also classify the noise to identify which area has more severe noise pollution and what the primary sources are. Although a manual approach may be used to gather environmental sound, it is more expensive and wasteful of human resources, especially when it is required to cover a large area

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for an extended period. An autonomous system using distributed IoT sensor nodes for long-term monitoring of noise distribution, levels, and sources is a good choice compared to the manual approach. However, to the best of our knowledge, there is often a trade-off between large-scale deployment, affordability, and classification ability in existing noise monitoring systems, resulting in the inability to achieve all three simultaneously. Therefore, we propose a distributed hierarchical wireless acoustic sensor network (WASN) based environmental noise monitoring system using a direct-connected wireless sensor network (WSN) and its extended version, a multi-hop wireless mesh sensor network (WMSN), for audio-level high data rate transmission. A single-board-computer-based central processing module in the proposed network receives audio for sound classification and A-weighted SPL measurement. The proposed system is targeted as low-power, low-cost, and highly scalable. (Fig. 1)

In this study, our main contributions are including:

- A distributed hierarchical WASN for environmental noise monitoring is proposed. To the best of our knowledge, no other research has developed a distributed WASN that transmits the high-speed audio data from end nodes to the central processing module for performing A-weighted SPL or doing sound classification.
- The Base64 algorithm is tested in the proposed system and verified insufficiently to increase the proposed network stability by decreasing the data throughput.
- A series of tests and comparisons in diagnosing the performance with respect to recording continuity, packet loss, recording quality, accuracy on A-weighted sound pressure level calculations, and costs are conducted to compare the performance of the proposed network.

The remainder of the paper is organised as follows. Studies related to the paper, including the weaknesses, are introduced in Section 2. Section 3 and Section 4 describe the proposed WASN in hardware and algorithm, respectively. The test results and discussions are contained in Section 5. Finally, Section 6 concludes the paper.

2. Related work

Many researchers have considered monitoring environmental noise using WSN, with most proposed approaches focusing on measuring SPL. In studies [8,9,10,11,12,13,14], various SPL measuring networks were developed based on different platforms (e.g. Raspberry Pi, Arduino, ESP8266, ESP32, etc.) and communication methods (e.g. wifi, GPRS, LoRaWAN, Bluetooth, etc.). The common idea behind these systems is to collect sound using the microphones, compute the SPL locally, and then upload the results directly to clouds or gateways (as shown in Fig. 2). The nodes operate independently, meaning they do not rely on centralised coordination and are not affected by neighbouring node failures. This approach significantly reduces communication requirements by transmitting only SPL results rather than raw audio data, thus lowering data volume. However, balancing communication range, power consumption, and system costs remains challenging with direct connection methods, as they typically require expensive and power-intensive transmission modules for broader deployment.

To address the issue, Wang et al. extended the communication range through ZigBee-powered WMSN, as shown in Fig. 2, which supports multi-hop communications. The end nodes powered by ZigBee are connected to each other, transmitting the SPL via the neighbour end nodes with a destination of a gateway that communicates to the data centre for further data processing [15]. Low-capability platforms, such as Arduino, ESP8266, and ESP32, are usually used as end-node processing units because they are inexpensive and have enough computational power for SPL calculations. Thus, using low-capability platforms in such projects decreases costs and power consumption compared to using high-capability platforms such as Raspberry Pi. Despite the advancements made by Wang et al. in extending coverage using multi-hop communications, all the mentioned systems suffer from a common drawback: they lack the functionality to determine sound sources. Consequently, these systems can only provide SPL and the corresponding geographical distribution without offering insights into noise sources. This functional limitation hinders local management teams from obtaining the critical information necessary for fast and accurate action.

Compared to SPL computation, sound source classification demands significantly higher computational power. Consequently, high-capability platforms are typically employed for this task, with the results uploaded to the cloud or gateway. Maijala et al. proposed a noise monitoring system that sampled and classified sound using Raspberry Pi-based nodes. Each node is equipped with a solar panel, a set of batteries, and a 3G/4G modem, enabling it to be freely deployed in areas where wired power and network access are not feasible [16]. Tan et al. proposed two types of systems utilising Raspberry Pi to classify noise. One type used a single microphone solely for noise classification, while the other employed a 7-microphone array for both classification and calculating the direction and distance of

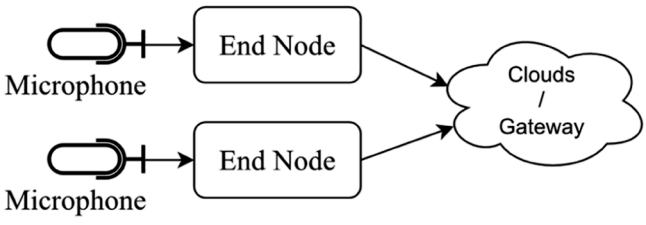


Fig. 1. Direct connection.

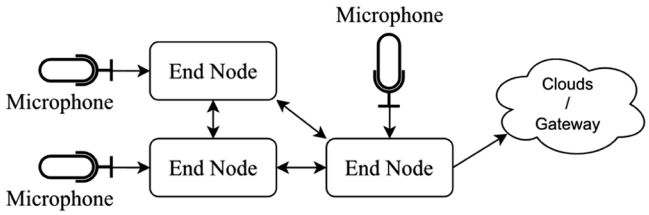


Fig. 2. Mesh connection.

the noise source [17]. To enhance classification accuracy, Vidaña-Vila et al. proposed a ring topology for multiple nodes deployed at the same traffic junction. In this configuration, each node performs classification simultaneously, ensuring that the uploaded results are corroborated by more than one node [18]. However, the limitations of the above projects are evident: each Raspberry Pi process sounds at a single location, restricting the monitoring range of each high-capability node. The only way to increase deployment density is by adding more Raspberry Pi in target areas, which is costly and power-intensive.

In summary, while existing noise monitoring systems that only measure SPL have proven capable of expanding coverage using multi-hop networks, they lack the functionality to provide noise source classification, limiting the value of these systems. On the other hand, current systems that are capable of sound classification perform the classification at each end node, where each high-capability device is connected to a single microphone or a microphone array. Thus, when expanding the coverage, each high-capability device only analyses noise at a single location, resulting in high power consumption, elevated costs, and limited scalability.

3. The proposed network structure

The proposed distributed hierarchical WASN for environmental noise monitoring system targets classify sound sources and measure A-weighted SPL. Considering that the existing systems, such as [16,17,18], did not make the best use of the high-capability platforms' performances for classification, the proposed network feeds data from multiple low-capability-platform-based sound recorders to increase the number of sound sources. The system aims to address the high power, high cost and low scalability issues in existing IoT-based systems that prevent the technology from being deployed and used.

3.1. Network topology

To achieve goals that are inexpensive and have a wide coverage area, the use of many low-capability platforms is the best option. However, unlike systems such as [9,11,13,14,19,20,21,22,23,24,25,26], where low-capability platforms process audio obtained by sensors, the computing power of the end nodes is not sufficient to support calculations like sound classification, meaning that the audio signal has to be transmitted out from end nodes to the central processing node with a reasonable data rate. Single-hop communication between end nodes (at the bottom tier) and the central processing module (at the top tier) is not a good choice in the proposed network, considering high-speed, low-power, long-range, and single-hop communication modules increase the system costs by deploying on each end node.

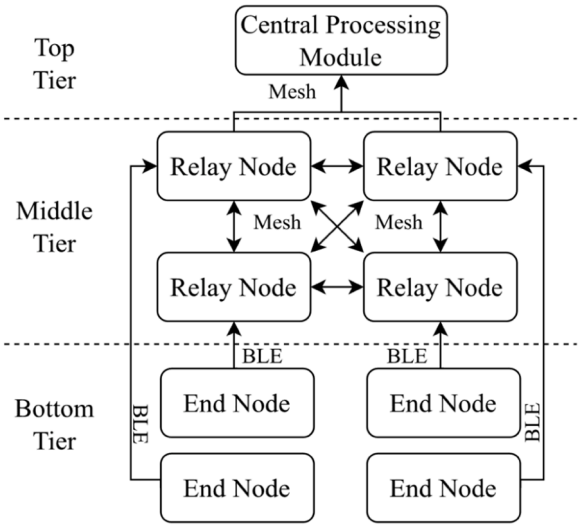


Fig. 3. Network topology.

In wide-range networks where nodes are deployed densely, multi-hop networks offer advantages over single-hop approaches in terms of reduced hardware costs by using high-speed, low-power, and short-range communication at each hop. Nodes between the end nodes and the central processing module are called relay nodes (at the middle tier), which receive data from end nodes and forward it to the central processing module. Multi-hop-based WMSN further extends the existing advantages with outstanding stability, scalability, and self-healing due to its ability to autonomously determine the strength of connections to switch routing routes so that it provides a reasonably low power design while satisfying the service quality requirements. Fig. 3 shows the proposed hierarchical WASN topology, which uses both the direct-connected WSN and the multi-hop WMSN. The bottom tier nodes transmit audio to the middle tier nodes through direct connections, and the middle tier nodes communicate with the central processing module via the WMSN they build together. The middle tier is the network's backbone, so the mesh network is expected to have a higher data rate and a broader communication range than the direct connections between relay nodes and end nodes. The central processing module is based on a high-capability platform to classify sound and compute SPL, whereas low-capability platforms should be applied to relay nodes and end nodes to decrease costs and power consumption.

3.2. Platforms and communications

The proposed environmental noise monitoring system structure uses a wifi mesh network (2.4 GHz) as the backbone for high-speed, flexible, scalable, and self-healing features. For the initial implementation, a node chain in one direction is shown in Fig. 4, based on the network topology, communication protocols, and hardware selection presented in previous sections. There are three types of platforms in the system. SLTB004As sample sound signal and transmit the audio to the corresponding ESP32 via BLE (solid blue arrow). The ESP32 forwards received data into the wifi mesh network (solid pink arrow) with the destination of Jetson Nano. According to the sound classification system proposed in our study [27], the time required to classify 6.144 s of audio on a Jetson Nano is 35 milliseconds. This implies that, in theory, each central process module can process data from up to 175 end nodes. The command flows are indicated as dotted arrows in Fig. 4. For instance, the Jetson Nano sends a command to enable sound sampling of the system to ESP32s and then forwards it to SLTB004As. However, due to the difficulties of sending commands directly from the Jetson Nano into the wifi mesh network, the commands must be delivered via wired-connected UART from the Jetson Nano to the bridge node, followed by sending into the mesh network. This difficulty is explained in detail in the following sections.

4. Network development

4.1. End node (SLTB004A)

End nodes based on SLTB004A (algorithm shown in Fig. 5) are designed to sample environmental noise and transmit the audio to relay nodes via BLE. The human ear is most sensitive to sounds in the frequency range of 3 to 4 kHz [28], and the environmental noise

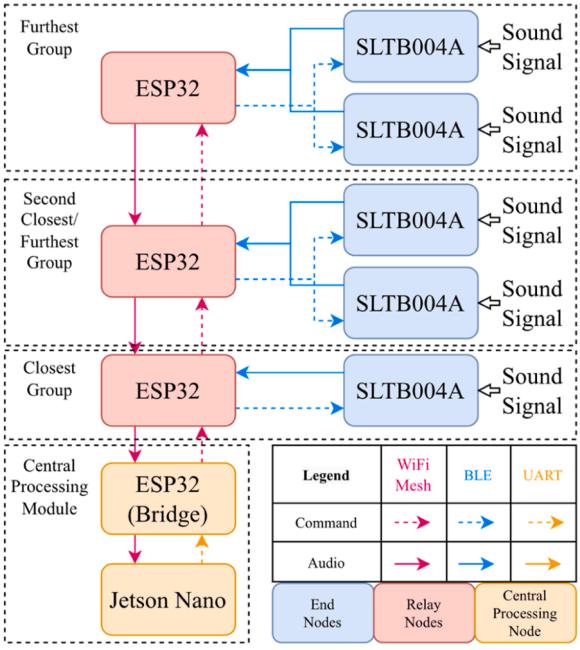


Fig. 4. One node-chain design based on the network topology.

that can have effects is mainly at frequencies below 4 kHz [29]. Thus, according to Nyquist's theorem, SLTB004a samples environmental sound at 8 kHz can cover the sound frequencies that have the most significant impact on people and decrease power consumption compared to samples at a higher rate. The microphone has a sample depth of 16 bits, making the sampled values lay in a range of −32,768 to 32,767. Thus, the microphone's output data rate is 128kbps (8kHz × 16 bits).

Since the network is based on low-capability platforms with limited computation power and restricted power consumption, audio compression techniques should be applied to reduce the data rate and improve transmission efficiency. ADPCM is a lossy audio coding algorithm with low latency and encoding complexities [30], making it advantageous for tasks where recording is done using low-capability platforms without excessive demands on sound quality. ADPCM encoder has a compression ratio of 4:1 since two 16-bit audio samples (4 bytes) can be encoded into one 8-bit integer (1 byte). Each time, 112 16-bit integers are compressed into 56 8-bit integers, and to make sure the data can be decompressed correctly, the 16-bit mean and maximum absolute values of the 112 integers are attached to the compressed array as four 8-bit integers. Hence, 112 16-bit integers are encoded into 60 8-bit integers and stored in a buffer until there are 240 8-bit integers ready to be transmitted as one packet to the relay node via BLE. Therefore, the audio data rate after encoding is decreased to around 34.29 kbps, i.e., about 4285.71 8-bit integers per second.

4.2. Relay node (ESP32)

A mesh network established by multiple relay nodes based on ESP32s is the backbone of the proposed environmental noise monitoring network, and the mesh network's stability is critical. The ESP32s are programmed using Arduino IDE, and PainlessMesh from the Arduino library establishes the wifi mesh network. PainlessMesh-based mesh network running on ESP32s has a millisecond-level delay and up to 9 milliseconds for a non-line-of-sight full-mesh network with two hops [31], which is low enough in the environmental noise monitoring systems. It is important to note that JSON object is used in PainlessMesh messaging to make the messages accessible and readable for developers, which means data in other types, an integer, for instance, should be converted to a string first, followed by embedding into a JSON object and send into the mesh network.

The algorithm in ESP32s is shown in Fig. 6, where the solid arrow indicates the audio flow, and the dotted arrow represents the command flow. The algorithm can be categorised into three streams. The first stream is data routing, where ESP32s need to route data from further ones to the next ESP32 until the message destination. This process is straightforward without any data conversion, but this stream involves both audio and command flow. The second stream is for the audio transmission, where the 8-bit integers obtained from the connected end nodes are converted to a string and embedded into a JSON object. We convert each 8-bit integer into two 8-bit characters because not all 8-bit integers have the corresponding printable characters in ASCII. For instance, a decimal integer with the value 127 is a control character and unprintable, so we convert it into "7F", which is the 127 in hexadecimal. The data rate is doubled to 68.58 kbps after the string conversion. Each packet transmitted from end nodes contains 240 integers, which are converted into a string and stored in the relay nodes for transmitting when there are three packets, which include 1440 8-bit characters (representing 720 8-bit integers). Additional information, such as the end node MAC address and packet count, are embedded into a JSON object with the audio to identify the data source. The third stream analyses JSON-object commands from the central processing module and sends the commands in integers to the end nodes via BLE.

4.3. Central processing module (ESP32 and Jetson Nano)

The central processing module contains two parts: a Jetson Nano and a bridge ESP32. They are expected to be deployed together and considered as one single node where the ESP32 is wired to connect to the Jetson Nano to get power and a UART connection. We connect the Jetson Nano with the wifi mesh network by starting a wifi hotspot on the Jetson Nano and programming the bridge node to connect to both the wifi hotspot and the mesh network. PainlessMeshBoost is a tool used in the Jetson Nano to make it a server node with a given node ID so that all relay nodes in the wifi mesh network sending messages to the node ID can be received by the Jetson Nano. However, PainlessMeshBoost only operates in the terminal, so we can only receive messages from relay nodes without the capability to send any. Therefore, there are two streams in the bridge ESP32 algorithm, as shown in Fig. 7. One stream is for data routing, which forwards the audio encoded as JSON objects from relay nodes to the Jetson Nano via a wifi connection. Another stream receives commands from the Jetson Nano via UART, embedding the command into a JSON object and broadcasting the message to all relay nodes.

The algorithm in Jetson Nano can be shown in Fig. 8. The JSON decoder analyses the JSON object to get information such as the

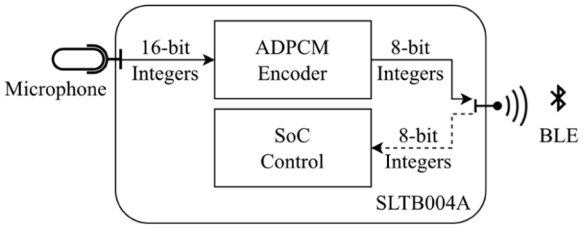


Fig. 5. Algorithm at SLTB004A.

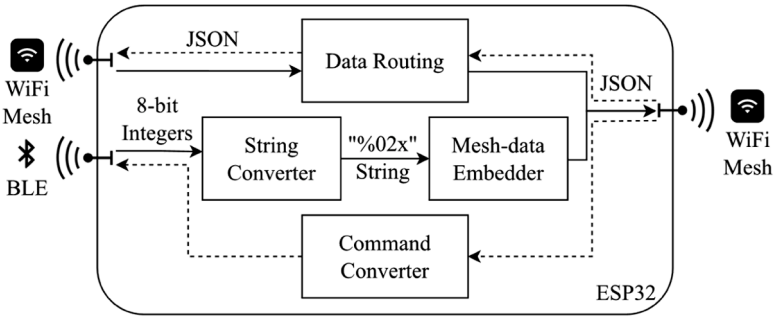


Fig. 6. Algorithm at ESP32.

message type, the source end node, the packet count, and the audio string. Then, the audio string is fed into the ADPCM decoder to decompress data packet by packet. The audio data distribution is restored following the equations:

$$X' = \frac{X - mean(X)}{\max(abs(X - mean(X)))}, \tag{1}$$

$$Y = X' \times (MA \pm Mean) + Mean, \tag{2}$$

where X is the decompressed audio, MA and Mean are the maximum absolute and mean values of the original audio, respectively, and Y is the final audio array. The calculation "±" depends on the data in X', where "+" is used when max(X') is smaller than abs(min(X')) and "-" is used in other cases.

A received packet counter is implemented to compare with the packet count number in the received packet to determine if there is a lost packet in the wifi mesh network and how many are lost. This network is going to do A-weighted SPL and classification based on the decompressed audio in the Jetson Nano, but since this paper focuses on the network, the classification algorithms are not included. Commands are generated from the Jetson Nano and transmitted to the bridge node via UART, followed by being sent into the wifi mesh network and BLE connections to control end nodes remotely. In this way, all end nodes in the system can start or stop sound sampling simultaneously, regardless of the breadth of deployment, as long as it is within communication range.

4.3. Base64 coding

In the network proposed in previous sections, we try to send audio converted into string in the format of "%02x" to the wifi mesh network. The conversion outputs two 8-bit characters from an 8-bit integer. The doubling data throughput may place a burden on the network, which may lead to connection interruptions or even system crashes. A commonly used method to include binary files, such as images in text-only documents, is encoding binary files using Base64 into text [32]. Suppose three 8-bit integers, A, B, and C, taking 24 bits in total, are trying to be represented as a string; the output contains six characters (48 bits) by using the "%02x" string converter; however, if we use Base64 encoder to group the 24 bits by six and represent the six-bit values by characters, four characters (32 bits) will be enough for representing three integers as texts. This approach can decrease the communication data throughput by 1/3. Thus, the 68.58-kbps audio-string data rate can be decreased to 45.72 kbps. Therefore, we implemented Base64 encoders at end nodes (Fig. 9) and relay nodes (Fig. 10) to compare different implementations' performances. The other components in the corresponding networks, such as the Jetson Nano, need to be changed to fit the Base64 encoding.

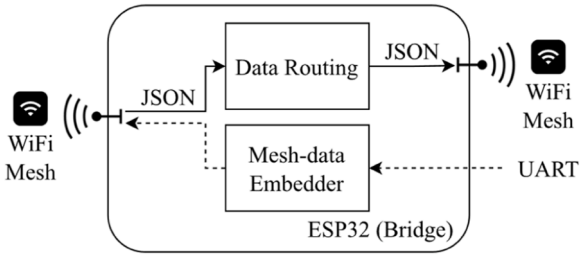


Fig. 7. Algorithm at the bridge ESP32.

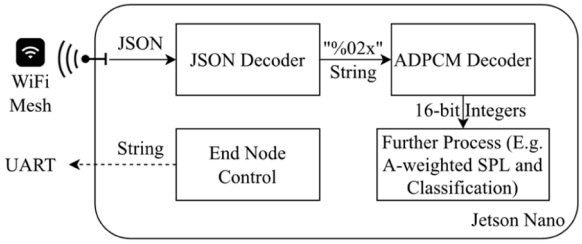


Fig. 8. . Algorithm at Jetson Nano.

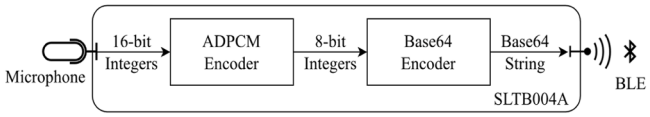


Fig. 9. Algorithm at SLTB004A with Base64.

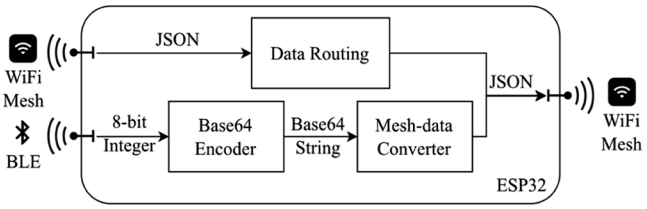


Fig. 10. Algorithm at ESP32 with Base64.

5. Test results and discussions

5.1. End node performance

End nodes in the proposed network are based on SLTB004As, a low-capability platform with a 38.4 MHz operating frequency [33]. Since the end node is expected to sample sound signals continuously, the workload should not be heavy to avoid the significant gap between each sampling period. Hence, measuring how long an end node takes to process a known period of audio to evaluate the algorithms is critical. There are 240 8-bit integers in each packet transmitted out from SLTB004As, which include around 56

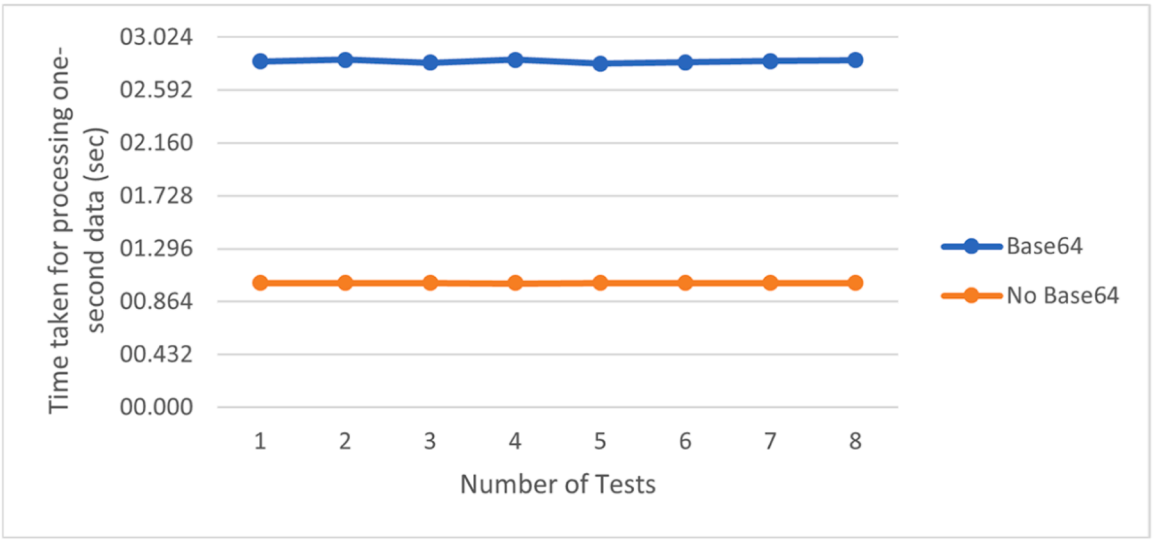


Fig. 11. Time for SLTB004A records and transmits one-second audio.

milliseconds at an 8 kHz sampling rate, and we monitor the time taken to send 358 packets (20.048 s) from end nodes with and without Base64. The reason for using 358 packets (20.048 s) rather than several packets is that we expect a value calculated on average to decrease any impact made by the added codes for this test. Eight tests were continuously delivered in the lab conditions. After calculating the time taken to process one-second audio, the test results are presented in Fig. 11, where the top and bottom lines indicate the end node with and without using Base64, respectively. The average time to record, process, and transmit one-second audio is 1.016 s without Base64, which is close to 1 second and has a processing and transmitting time of 99.12 % ($\frac{(2.826-1)-(1.016-1)}{2.826-1}$) lower than using Base64 (2.826 s). Additionally, with a close look at the figure, the bottom line is almost straight, and the top line has more fluctuations, so the algorithm with Base64 has greater instability. Thus, we prefer to use the algorithm without Base64 at end nodes.

5.2. Packet loss in the network

The network test processes are performed based on the entire network setup in the lab conditions, where each direct connection has a 5-meter distance (shown in Fig. 12, where the end-node MAC addresses are included under the devices). The block diagram for this experiment is shown in Fig. 4. In the proposed network, the wifi mesh is expected to transmit messages with a data rate of 68.58 kbps (without Base64) or 45.72 kbps (with Base64) for each end node and the BLE is expected to transmit at 34.29 kbps, which are significantly heavier than existing systems that transmit results only. To test the BLE connection, SLTB004As transmit each packet (240 8-bit integers, around 56 milliseconds) with a packet ID, a counter value to indicate how many packets have been sent out. The corresponding ESP32s receive packets, comparing packet IDs with receive-packet counters to monitor if there is a packet loss in BLE transmission. The network continuously runs for 24 h, and according to our test, no packet was lost during the BLE transmissions under the introduced conditions.

Compared to BLE, wifi mesh needs to handle higher data throughput, and the hops closer to the Jetson Nano take heavier throughput than further ones. Similar to the method introduced in Section 5.2.1, each packet sent from relay nodes has a packet ID and a self-increment counter indicating the number of packets transmitted. At Jetson Nano, five counters (one for each end node) store the number of received packets compared to the packet IDs in the received packets to identify if there is a packet loss. Each packet we transmit contains 720 8-bit ADPCM-compressed audio, which is 0.168 s. Relay-node algorithms with and without Base64 are tested to determine whether the Base64 encoding process improves the network stability by decreasing the data throughput or simply increasing the computation pressure to relay nodes. Each algorithm has been tested by continuously running for 12 h. We further analyse the results to obtain the packet loss rate over time by dividing the number of lost packets by the number of packets. The wifi-mesh-test results are shown in Fig. 13, where (A) and (B) use the algorithm without Base64, and (C) and (D) show the algorithm using Base64. Note that the end nodes are aligned based on the distance to the central processing modules, i.e. end node 84:fd:27:e6:5c:f1 is closest to the central processing module, which is consistent, as Fig. 12 shows.

During the 12-hour running test on the algorithm without Base64, only three lost packets came from end node 04:cd:15:72:fc:b8,

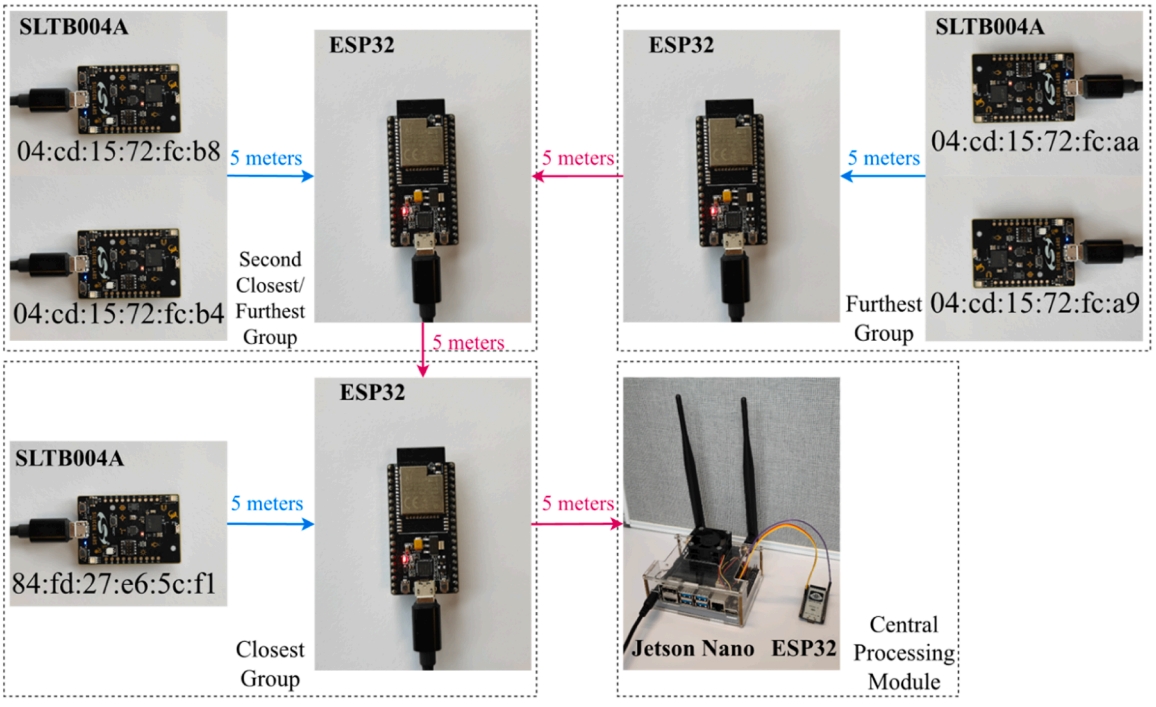


Fig. 12. Network testing scenario.

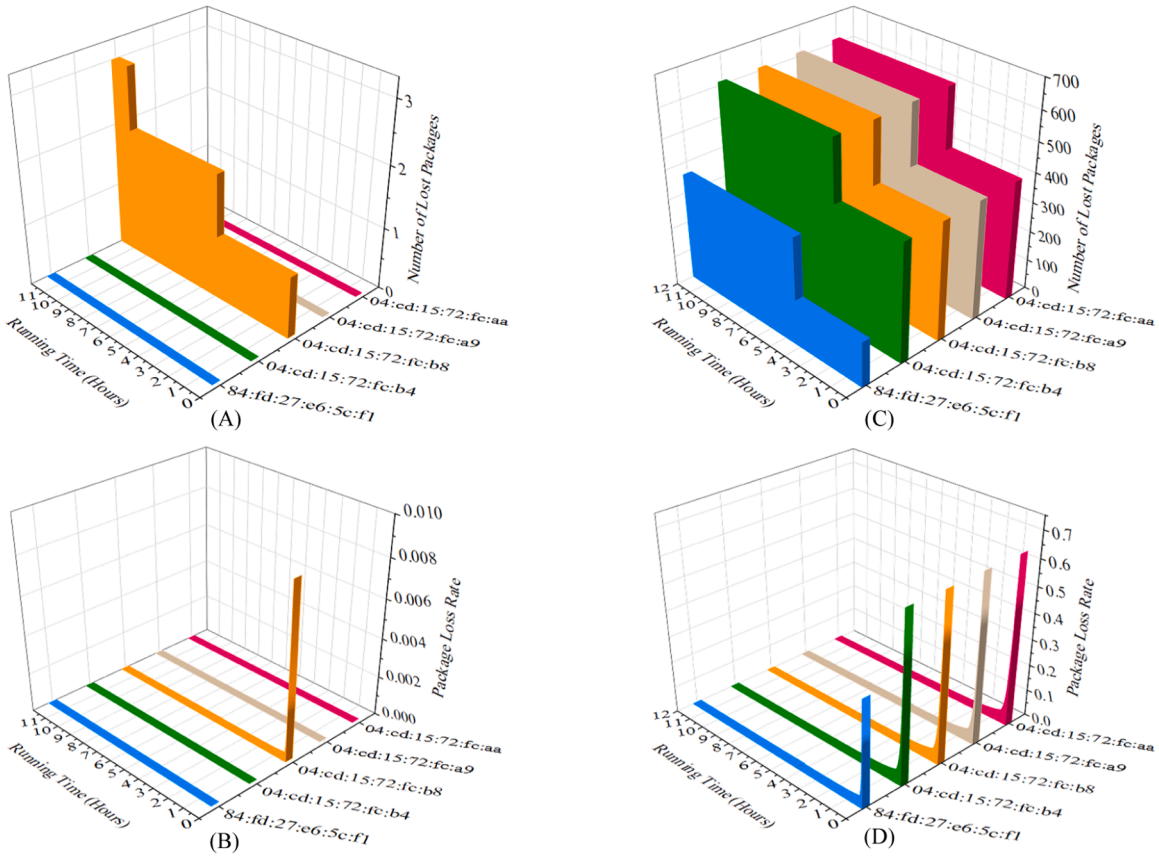


Fig. 13. (A) Number of lost packages without using Base64 over time. (B) Package loss rate without using Base64 over time. (C) Number of lost packages using Base64 over time. (D) Package loss rate using Base64 over time.

and the time intervals between packet losses were approximately 5 h. The packet loss rate at this end node approaches zero as the number of packets increases, while the packet loss rates are zero at other end nodes. Three packet losses lead to a 0.504-second (3 packets with 0.168 s per packet) loss in 12 h, which is a satisfactory result. There are significantly more lost packets using the algorithm with Base64 in the 12-hour test. Each end node has 362,611,610,613, and 613, respectively, from the closest end node to the furthest at the end of the test. During the test, we found that no crash occurred at any node or error at the bridge. Considering there is no packet loss in BLE, the wifi mesh instability must cause the packet loss before the bridge. The instability affects data transmission from further relay nodes that count on the failing one.

According to Fig. 13(C), the furthest and the second furthest relay nodes have very close packet loss across the whole test (around 610 to 613 packets at the end), which means their connection status to the central processing module is consistent with all the time. Following the same idea, the closest and the second closest (or second furthest) relay nodes have unstable connections because of the significant gap caused by the disconnection between the two nodes. This disconnection happens at 0.0295 h (106 s) after the test starts because the 250-packet gap occurs at this moment and lasts until the end. Finally, the closest relay node disconnects with the bridge twice, and each time leads to packet loss at all end nodes. Thus, the closest hop is disconnected twice, the second closest hop is disconnected once, and the furthest hop is never disconnected, which concurs with our expectation that closer relay nodes need to process more data, leading to frequent instability. Although the packet loss rate is diluted and close to zero while the packet amount increases according to Fig. 13(D), the network lost 2809 packets in 12 h, which is a 471.912-second audio.

In conclusion, although Base64 is expected to decrease the data throughput in the wifi mesh network, the implemented Base64 algorithm is so heavy that low-capability ESP32s cannot have enough time to work on other tasks, such as communication maintenance, and can also cause device crash. According to the test results, the algorithm with Base64 lost 936.33 times more packets than without it. Therefore, the network without Base64 has the best stability and is used for further tests.

5.3. Recording quality

In this section, we compare the network-recorded with phone-recorded waveforms and spectrograms to verify the system-recording quality. At the end node, an audio signal is sampled with the microphone at a 16-bit 8 kHz sampling rate, which gives the bitrate at 128 kbps, followed by compressing the audio to 34.29 kbps with ADPCM encoder and wirelessly transmitting to the relay node via BLE. The

relay node expends data to 68.58 kbps and sends it to the Jetson Nano, which decompresses the received data with an ADPCM decoder for further processes. From this comparison, we expect the microphone's recording quality and the ADPCM algorithm's reproduction capability to be verified as acceptable.

In our lab, we record the voice of "one two three four five six" in English using the SLTB004A, which eventually reaches the Jetson Nano via the mesh network, and it is decompressed to produce the audio in WAV format for playback comparison. Simultaneously, we record the sound using a smartphone and resample the audio to 8 kHz. Both audios are normalised with the following equation

$$Y = \frac{X}{\max(\mathrm{abs}(X))}, \tag{3}$$

where X is the raw audio sequence, and Y is the normalised audio sequence. The waveforms and the spectrograms are shown in Fig. 14. The waveform recorded by SLTB004A shown in Fig. 14(C) has a very similar shape as the phone-recorded waveform shown in Fig. 14 (A) in general speaking. With a close look at the spectrograms (Fig. 14(B) and (D)), the SLTB-recorded audio has more power in high-frequency dark areas than the phone-recorded version, which means it may contain more high-frequency noise. This flaw is confirmed by converting the waveforms to WAV audio files, where the SLTB004A recorded audio is slightly noisier than the phone recording, although the words can be recognised easily.

Several factors may cause this difference in quality: firstly, as an IoT device, the SLTB004A has a microphone that needs to take into account energy consumption and therefore, the recording quality is not as good as the phone; secondly, recording applications on phones have more powerful filters that can accurately filter out background noise to retain pure vocals; last but not least, the lossy ADPCM encoder used in the SLTB004A compresses the audio with a loss of sound quality, and therefore the sound quality cannot be restored after decompression.

5.4. A-weighted SPL

To further verify that the proposed system can accurately sample environmental noise for further processes, including noise classification at the central processing node, we test the network's A-weighted SPL calculation by comparing the value with readings given by a sound level meter, CEM's DT-8851. This sound level meter gives A-weighted SPL between 30 dBA and 130 dBA with accuracy at ±1.4 dBA and a frequency range between 31.5 Hz and 8 kHz. The sound level meter and an end node (as shown in Fig. 15) are placed as close as each other to reduce the effect of sound transmission distance on the measured A-weighted SPL values.

We delivered the test in an office with uncontrollable sounds from air conditioners, traffic or helicopters outside the window, computer fans, colleagues' activities, etc., where the sound levels are generally around 40 dBA. For the test to cover high SPL scenarios, a speaker playing white Gaussian noise (WGN) is deployed at the same distance to both the sound level meter and the end node. We test

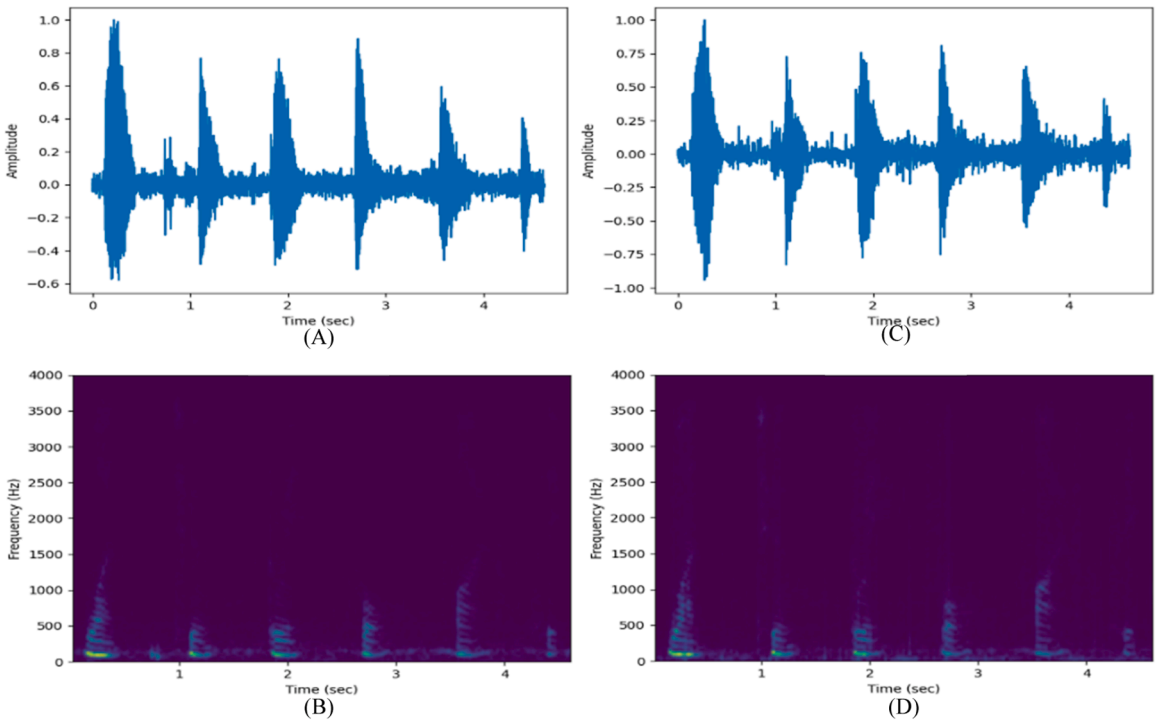


Fig. 14. (A) Phone-recorded audio waveform. (B) Phone-recorded audio spectrogram. (C) SLTB004A-recorded audio waveform. (D) SLTB004A-recorded audio spectrogram.



Fig. 15. A-weighted SPL test.

the system's accuracy in calculating different A-weighted SPLs by adjusting the volume level of the speakers.

Fig. 16(A) shows 325-second test results, where the blue dash boxes indicate the no-speaker scenario, and the orange dash boxes indicate the scenario with the speaker. Despite around ±2 dBA volatilities in the proposed network's A-weighted SPL results, it coincides with the trend of the results measured by the sound pressure meter in both low- and high-level areas. The data distribution shown in Fig. 16(B) can roughly reflect the accuracy of the A-weighted SPL calculations, where the proposed network and the sound level meter have measurement differences in the maximum value at 0.86457 dBA, the upper quartile at 1.04772 dBA, the mean at 0.53566 dBA, the median at 0.56782 dBA, the lower quartile at 0.35943 dBA, and the minimum value at 1.0431 dBA. Considering the sound level meter has an accuracy of ±1.4 dBA, which is higher than any of the five values, it is confident to say that the proposed network has excellent A-weighted SPL calculational performance.

5.5. Cost evaluations

Expensiveness is one of the weaknesses in existing environmental noise monitoring systems that the proposed network tries to address. We count the expenses of related work implementations according to their papers. As our test scenario allows for the simultaneous recording of five sources, we have calculated the total price of different implementations containing five sources to facilitate comparison with ours.

As the costs shown in Table 1 (our indicative cost of the components at the time of setting up the test is in bold), the cost of existing systems for processing both SPL and sound classification is approximately double that of the proposed network, being 170.00 %, 77.34 %, and 148.28 % higher, respectively. Compared to systems that measure only SPL, the proposed network has similar or even lower costs than three out of seven systems. While four systems have a clear cost advantage over the proposed network, they lack the crucial functionality of noise classification, giving our system a functional advantage. It is worth noting that the table is based on monitoring 5 locations only, but considering the Jetson Nano can classify sound from up to 175 nodes using our proposed classification system [27], the scale of the proposed system can be much larger than just 5 locations. As a result, systems that appear cheaper than ours in Table 1 will lose their cost advantage. Additionally, since some related studies were completed several years ago and those devices have become more affordable in today's market, the system prices were likely higher at that time.

5.6. Limitations

While our study demonstrates the effectiveness of the proposed WASN system, several limitations need to be addressed in future studies. Firstly, in the implementation of the proposed hierarchical network structure, ESP32 and SLTB004A are used as the relay nodes and end nodes, respectively. One or more SLTB004A devices can connect to an ESP32 via BLE to transmit sampled audio. The maximum number of simultaneous connections is 9; however, for stable performance in most applications, it is recommended to limit to 3 connections [49]. This means each relay node should not be connected to >3 end nodes to maintain optimal functionality and avoid potential issues. Secondly, the tests in this study were all conducted in a laboratory environment. Further studies will need to be carried out in more realistic conditions, such as weather variations or radio interference. Finally, the datasets used for testing the recording quality and A-weighted SPL were relatively small, with around 5 s and 320 s, respectively. In our future work, we plan to conduct more tests with larger datasets. Future works should address these limitations and explore the system's scalability in diverse environmental settings.

6. Conclusion

This paper proposed a novel environmental noise monitoring system structure based on a distributed high-speed hierarchical

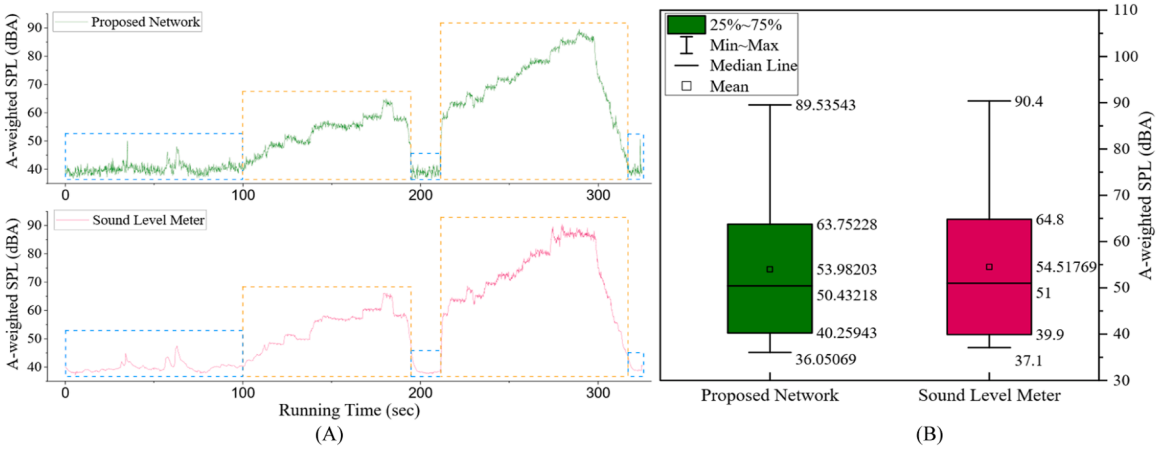


Fig. 16. (A) A-weighted SPL measurement. (B) Data distribution.

Table 1
Related work costs for monitoring 5 locations, including our proposal.

Studies	WASN-based systems' output types	Platforms (roles, costs)	Additional components (costs)	Total costs	Cost differences
Ours	**SPL and Sound classification**	**Jetson Nano (central processing module, US$149** [34]**)** **ESP32 (bridge and relay nodes, US$5)** **SLTB004A (end nodes, US$19.99** [35]**)**	Intel 8265 wireless card (US$12.99 [36])	**US$281.94**	**Baseline**
[8]	SPL	Raspberry Pi (end nodes, US$25 [37])	UA0053 sound card (US$16.77 [38]) Microphone (US$0.78 [39])	US$212.75	−24.54 %
[9]	SPL	Arduino ATMega 2560 (end nodes, US $48.4 [40]) ESP8266 (end nodes, US$7.5 [41])	Sound sensor	US$279.5	−0.87 %
[10]	SPL	Raspberry Pi 3B (end nodes, US$35 [42])	LM393 sound sensor (US$0.12 [43])	US$175.6	−37.72 %
[11]	SPL	Arduino Uno Rev3 (end nodes, €22 [11])	Lo-Ra click module (€54 [11]) Grove sound sensor (€5 [11])	€405 ≈ US$425.25	+50.83 %
[12]	SPL	Raspberry Pi 2B (end nodes, US$35 [44])	Sound sensor	US$175	−37.93 %
[13]	SPL	ESP8266 (end nodes, US$7.5 [41])	LM393 sound sensor (US$0.12 [43]))	US$38.1	−86.49 %
[14]	SPL	Arduino Due (end nodes, US$48.4 [45])	Adafruit Electret Microphone (U.S. $6.95 [46]) Communication components	US$276.75	−1.84 %
[16]	SPL and Sound classification	Raspberry Pi (end nodes, US$25 [37])	Microphone 3 G/4 G module Other components	€150 [16]×5 = €725 ≈ US$761.25	+170.00 %
[18]	SPL and Sound classification	Raspberry Pi 2B (end nodes, US$35 [44])	OUT-AMLO-0872 microphone (€12 [18]) Communication components	US$100 [18]×5 ≈ US$500	+77.34 %
[17]	SPL, Sound classification, and Position	Raspberry Pi 3B+ (end nodes, US$35 [47])	UMA-8 USB mic array (US$105 [48]) Other components	US$700	+148.28 %

* Cost difference = (Total cost – Baseline cost) / Baseline cost × 100 %.
* €1 ≈ US$1.05.

WASN for sound transmission and classification purposes. We addressed and implemented solutions to the essential issues in the design of such systems, such as power consumption, costs, and scalability. The network comprises three layers: the bottom layer contains a number of low-capability end nodes (SLTB004A) to sample environmental sound and send the audio to nodes located in the second layer via BLE. The second low-capability relay nodes (ESP32) layer forms a wifi mesh network for multi-hop communication so that the audio can be relayed from the farthest nodes to the cluster head. The top high-capability node (Jetson Nano) layer receives data from the mesh network for further processing.

We tested the network's initial implementation with three relay nodes where data from the farthest relay node needed three hops to reach the cluster head. Each relay node gathers audio from one, two, and two end nodes depending on the required number of hops from one to three. Executed tests showed that the end nodes had significantly low delays in the recording and transmitting process with 1.016 s processing time for one-second audio. Our network had excellent stability, with a 0.504-second audio loss in 12 h of continuous

recording. By comparing the phone- and network-recorded waveforms and spectrograms, the proposed network was verified to have excellent recording quality. We also found that the A-weighted SPL delivered by the network fitted the data given by the SPL meter. Therefore, the system can sample environmental noise accurately at multiple locations simultaneously for further processes, such as noise classification. The proposed network showed excellent performance under several test scenarios and a cost advantage compared to existing rivals.

CRediT authorship contribution statement

Bo Peng: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Resources, Software, Validation, Visualization, Writing – original draft, Writing – review & editing. **Kevin I-Kai Wang:** Supervision, Writing – review & editing. **Waleed H. Abdulla:** Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

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Data availability

Data will be made available on request.

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